

Paper Title & Reference	Core Approach	Key Implementations	Identified Gaps / Limitations	Contribution of My Project
<i>"The Urban Vision Hackathon: Image Annotations for Indian Traffic"</i> (arXiv 2024)	CV (YOLO)	Large-scale 26k image dataset (UVH-26) for Bengaluru traffic.	Focused purely on detection; no execution or scheduling logic.	Execution Layer: Integrates UVH-26 weights into a live OS-driven control system.
<i>"Real-time Traffic Light Control Based on CPU Process Scheduling"</i> (IEEE ASID 2011)	OS Theory	Mapping vehicle flow to CPU scheduling (RR, SJF).	Relies on manual/static sensors; no AI-based perception.	Automated Perception: Uses YOLO CV to automate the sensing of vehicle "processes."
<i>"Deep Reinforcement Learning for Adaptive Traffic Signal Control"</i> (IEEE ITS 2023)	AI (DRL)	Uses RL to optimize signal timing in real-time.	"Black Box" logic; difficult to debug or explain in safety-critical use.	Explainable AI: Uses deterministic OS algorithms (MLFQ) that are easily auditable.
<i>"Digital Twin Intersection: A Framework for Urban Planning"</i> (Smart Cities 2023)	Digital Twin	Created a static virtual replica of city intersections.	Non-reactive; does not adapt to live video or surges.	Dynamic Twin: Created a "Live Twin" that reacts frame-by-frame to traffic_video.mp4.
<i>"Vehicle Counting and Classification using YOLOv8 in ITS"</i> (IEEE Access 2024)	CV (YOLO)	High-precision vehicle counting on standard highways.	Does not address lane starvation or fairness under imbalance.	Fairness Engineering: Implements MLFQ with Aging to prevent lane starvation.
<i>"Emergency Vehicle Preemption using Infrared Sensors"</i> (IEEE 2023)	Hardware	Hard-coded priority for emergency vehicles.	Binary "Green/Red" logic; causes massive congestion in side lanes.	Smooth Priority: Uses priority queues within MLFQ to handle ambulances without gridlock.

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<i>"Indian Driving Dataset: A Dataset for Heterogeneous Traffic"</i> (IDD 2022)	CV (Sensing)	Dataset for self-driving cars on unorganized Indian roads.	Focused on ego-vehicle navigation, not infrastructure control.	Infrastructure Focus: Uses Indian context for infrastructure-level signal management.
<i>"Modified Round Robin for Dynamic Traffic Management"</i> (IEEE 2022)	OS Theory	Dynamic time-slicing based on lane load.	Cannot handle highly unbalanced lanes (Starvation occurs).	Priority Multi-leveling: Uses 4 priority levels (MLFQ) to manage extreme lane imbalance.
<i>"Forecasting Traffic Flow via Graph Neural Networks"</i> (GNN 2024)	ML (GNN)	Predicting traffic patterns using historical network data.	Fails during "Black Swan" events or novel traffic incidents.	Instant Perception: Reacts to live arrivals via CV, requiring zero historical training data.
<i>"Pedestrian Safety in Adaptive Traffic Control Systems"</i> (IEEE 2023)	Algorithmic	Dynamic priority for pedestrian and bike safety.	Lacks a "Stress Test" mechanism to validate under heavy load.	Experimental Rigor: Uses the Multiplier Trick to stress-test safety logic under heavy pressure.